Agent Colosseum

Event PRD v2

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1. Executive Summary & Event Concept

Agent Colosseum is a multi-stage, gamified hackathon where teams build and deploy autonomous AI agents using standard APIs, custom tool pipelines, standard Python libraries, and framework-agnostic LLM integrations (LangChain, LlamaIndex, direct REST calls, or custom agent loops).

Unlike traditional static hackathons, Agent Colosseum operates like a real-time strategic game:

1. Teams select a **domain track** from the fixed set offered at the event.
2. Teams write their **own problem statement** within that track and submit it for approval during onboarding.
3. Teams receive **two generic domain tasks**, revealed at **10:00 AM**, Task 1 first. The tasks apply to any problem statement within the track.
4. Teams start with a balance of **1,000 Colosseum Credits (CC)** and **spend** them, selecting tasks and purchasing components from the Feature Store to assemble their agent.
5. Teams may enter the **Game Arena** at any point to earn additional credits through mini-games. Entry is optional and open to every team from the start, not a fallback for teams who have run out.
6. Teams wager remaining credits in **Casino Royale** for score multipliers or model perks.
7. Laptops close for the final **Automated Colosseum Arena**, where agents are stress-tested zero-touch live on stage.

Credits count down toward a **minimum threshold of 100 CC**. The objective is to finish as close to that threshold as possible while still building a complete agent. Credit discipline is scored, so an unspent balance carries a cost and reckless spending that leaves nothing to wager carries its own.

There is no starter repository. Teams begin from an empty repo of their own and construct the agent from Feature Store components and their own code.

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2. Master Event Schedule & Phase Workflow

Total Duration: 3 Hours 30 Minutes

MASTER EVENT FLOW				
PHASE 0 Onboarding (15 Mins) Track & PS Selection	PHASE 1 Integration (45 Mins) Task 1 + Store Spending	PHASE 2 Orchestration (60 Mins) Task 2, Chaos & Game Arena	PHASE 3 Casino Royale (20 Mins) Wager Credits For Multiplier	PHASE 4 Gauntlet (45 Mins) Zero-Touch Live Test

Phase Breakdown

Phase ID	Stage Name	Duration	Platform State	Participant Objective	Transition Trigger
PHASE_0	Onboarding & Briefing	15 Mins	Onboarding Unlocked	Log into platform, receive team credentials, select a domain track, submit a problem statement for approval, receive the 1,000 CC opening balance, verify local environment.	Admin pushes Phase 1 transition via `/admin`.
PHASE_1	Task 1: Integration	45 Mins	Task 1 Revealed, Feature Store Open,	Task 1 is revealed at 10:00 AM. Teams pay the selection cost (30 to 50 CC) to unlock it, then purchase Feature Store components and wire the domain data sources and services their agent requires. Pass 3	Phase 1 timer hits 00:00 or Admin

			Game Arena Open	verification tests.	triggers Phase 2.
PHASE_2	Task 2: Orchestration & Chaos	60 Mins	Task 2 Revealed, Store Open, Game Arena Open	Task 2 is revealed. Teams build multi-step agent orchestration over their Task 1 integrations while handling live schema drifts and corrupted inputs. The Game Arena remains open to all teams.	Phase 2 timer hits 00:00.
PHASE_3	Casino Royale	20 Mins	Casino Open, Code Paused	Hands off code. Wager remaining credits on risk tiers to win score multipliers, API perks, or safe-haven locks.	All teams lock wagers / Admin triggers Phase 4.
PHASE_4	Colosseum Gauntlet	45 Mins	Submission & Arena Open	Submit final repo or drive link and lock entry. Zero-touch automated test suite bombards agents live on stage.	Automated evaluation run finishes.
PHASE_5	Podium & Wrap-Up	25 Mins	Read-Only	Final score calculation, organizer highlights, winner announcements, trophy distribution.	Event conclusion.

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3. Domain Tracks & Task Architecture

The Problem Statement Rule

Organizers fix the tracks. Participants choose their own problem statement within their track.

Each team submits a problem statement during PHASE_0. Technical Mentors approve or reject it against three criteria: it sits inside the chosen track's domain, it is achievable by an agent within the event duration, and it can be meaningfully addressed by both generic tasks.

The Two Generic Tasks

Every team within a track receives the same two tasks. Tasks are written at the domain level so they apply to any approved problem statement in that track.

```
                      STANDARD TASK FRAMEWORK

• TASK 1 (Integration): Connect the domain data sources, services and tools
  the agent requires. Establish working, verifiable I/O against each.

• TASK 2 (Orchestration): Build multi-step agent reasoning across those
  integrations, under degraded conditions: schema drift, corrupt inputs,
  inconsistent formats.

• PHASE 4 (Gauntlet): Live stream of 10 adversarial payloads (prompt
  injections, rate limits, corrupt schemas) evaluated zero-touch.
```

Worked Example: FinTech Track

Task 1 (Integration). Integrate the account and transaction data sources your agent needs. Your agent must authenticate against each source, retrieve records, and normalise them into a single internal representation. Verified by three tests: successful retrieval, correct normalisation of a known record, and graceful handling of an empty result.

Task 2 (Orchestration). Build a multi-step agent that reasons across those integrated sources to serve your problem statement, while transaction field names shift mid-run and a percentage of records arrive malformed. The agent must complete its reasoning chain without human intervention.

Both tasks hold whether a team's problem statement is fraud detection, ledger reconciliation, or spend categorisation. That is the test any track's task pair must pass.

Track Structure (Configurable per Event)

- **Track A (FinTech):** Transaction analysis, fraud detection, ledger reconciliation agents.
- **Track B (CyberSec):** Threat intelligence, log parsing, breach response agents.
- **Track C (Logistics / HealthTech):** Resource allocation, triage routing, dispatch agents.
- **Track D (Custom / Open):** Domain supplied by track sponsors or organizers.

Volunteer Deliverable Template (per track)

Each track owner submits:

1. Track name.
2. Task 1, phrased for the domain, with subtasks.
3. Task 2, phrased for the domain, with subtasks and the specific chaos conditions.
4. Acceptance criteria for each task that hold across any problem statement in the track.
5. The 3 Task 1 verification tests.
6. The 10 Phase 4 adversarial payloads, each with a pass/fail condition.
7. Feature Store components specific to the track, with proposed CC costs.
8. Proof of viability: both tasks applied against two different sample problem statements in the track.

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4. The Feature Store (Marketplace)

With no starter repository, the Feature Store is the primary build mechanism. Components purchased here are how the agent gets assembled. It opens at PHASE_1 and remains open through PHASE_2.

Credit Workflow

Event	Balance
Opening balance issued at PHASE_0	1,000 CC
Task 1 selection cost	-40 CC (range: 30 to 50)
Balance entering the store	960 CC
Component purchases through PHASE_1 and PHASE_2	drawn down from 960 CC
Game Arena earnings, at any point	added back

Teams spend down toward the floor. A large unspent balance at PHASE_3 indicates an under-built agent, not a lead.

Inventory Categories

```
                      ┌──► [ TOOL MODULES ] ──────────► Core agent capabilities
[ FEATURE STORE ] ────┼──► [ DEFENSIVE BUFFS ] ─────► Resilience against Phase 2 chaos
                      └──► [ OFFENSIVE SABOTAGE ] ──► Handicaps applied to rival teams
```

Illustrative Inventory

Specific items and costs are set per track by the track owner. The table below is illustrative and pending volunteer input.

Item Name	Category	Cost (CC)	Functional Effect	Status
Python REPL Sandbox	Tool Module	400 CC	Grants the agent the ability to write and execute arbitrary Python for dynamic computation and parsing.	Illustrative
Vector Memory Engine	Tool Module	350 CC	Adds vector store persistence to maintain state across multi-turn queries.	Illustrative
Air-Gap Guardrail Shield	Defensive Buff	300 CC	Strips prompt injection tags, system overrides and dangerous payloads before they reach the LLM.	Illustrative
Schema Inspector Tool	Defensive Buff	250 CC	Inspects source metadata prior to execution, granting immunity to dynamic column name shifts.	Illustrative

Prompt-Poison Injection	Sabotage	350 CC	Injects a trap payload into a targeted rival team's Task 2 input stream.	Illustrative
Network Lag Spike	Sabotage	200 CC	Applies a 3-second delay penalty to a target rival's API tool calls for 10 minutes.	Illustrative

Balancing Requirement

The store must be priced so that a competent, focused build is affordable within 1,000 CC without arena earnings, while an ambitious build with full defensive coverage and sabotage is not. Track owners price their components against this constraint.

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5. The Game Arena

The Game Arena is an optional credit earning station, open from PHASE_1 through the end of PHASE_2. It runs as a staffed physical station off the main floor.

Access

- Open to **every team at any time** while it is running. There is no balance requirement and no gate.
- Entry is capped at **4 runs per team** across the event.
- One team member plays at a time. The rest of the team keeps building.
- Teams may use the arena to fund a bigger build, to recover from overspending, or not at all.

Mini-Games

Three to five games, each 1 to 3 minutes, requiring no laptop. Specific games are pending selection. Selection criteria: fast to explain, fast to judge, no domain knowledge advantage, and playable repeatedly without becoming trivially exploitable.

Payouts

Payout per successful run should sit at roughly one small component's cost, so that a team can recover from a miscalculation but cannot fund a full build by grinding. Exact figures are set once the games are chosen and priced against the final store table.

Cost of Entry

The arena costs time, not credits. A member at the arena is a member not building. This is the intended self-limiting mechanism and should not be softened.

Staffing

One **Arena Operator** runs the station, verifies eligibility against the live balance, judges outcomes and credits winnings via the admin portal.

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6. Casino Royale

During PHASE_3, coding is paused. All teams wager remaining credits on the auditorium stage projector using three risk tiers.

```
                                 ┌──► [ THE VAULT ] ──────────► 100% Safe | 0% Bonus/Penalty
                                 │
[ CASINO TERMINAL ] ─────────────┼──► [ THE OVERCLOCK ] ──────► 50% Win: Tier-1 API Key
                                 │                              50% Loss: 3s Tool Lag
                                 │
                                 └──► [ THE HIGH-ROLLER ] ────► 30% Win: 2.5x Score Multiplier
                                                                70% Loss: -35% Total CC
```

1. **The Vault (0% Risk).** Locks in the current credit balance without applying multipliers or penalties.

2. **The Overclock (50/50 Coin Flip).** Wager: fixed 200 CC. Win (50%): unlocks high-throughput Tier-1 model keys for Phase 4. Loss (50%): applies a 3-second delay penalty to all tool executions during Phase 4.
3. **The High-Roller Bet (30% Win / 70% Loss).** Wager: 35% of the team's current balance. Win (30%): grants a 2.5x multiplier on all points earned during Phase 4. Loss (70%): deducts 35% of total credits immediately.

Bust Protection Rule: a team's credit balance can never drop below 300 CC from casino losses.

Note on wagering against a countdown balance: teams that have spent down close to the threshold arrive at PHASE_3 with little left to wager, and the Bust Protection floor means they risk little by trying. Teams holding a large balance can wager heavily but are already carrying a credit discipline penalty. This trade-off is deliberate.

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7. Scoring Logic & Final Leaderboard

Final rankings are calculated automatically upon completion of Phase 4.

Master Scoring Formula

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Final Score = (Gauntlet Points x Casino Multiplier) + Credit Discipline Score
```

Credit Discipline Score

Credits count down toward a minimum threshold of 100 CC. Teams are scored on how close their final balance sits to that threshold, so leaving credits unspent costs points.

```
Let  B = final credit balance at end of PHASE_3
     T = minimum threshold = 100 CC
     S = starting balance  = 1,000 CC

Credit Discipline Score = 150 x (1 - (B - T) / (S - T))
```

- A team finishing exactly at the threshold scores the full 150 points.
- A team finishing with 550 CC scores 75 points.
- A team that never spends scores 0.
- The result is floored at 0. A balance below the threshold, which can only occur through Casino losses under Bust Protection, scores 150 and no more.

The score is capped at 150 against a 1,000-point gauntlet maximum. It rewards spending the budget without ever letting credit management outweigh the agent itself.

Phase 4 Gauntlet Evaluation Metrics (1,000 Points Max)

- **Accuracy & Output Validity (40%):** percentage of test payloads returning valid outputs matching the required schema.
- **Adversarial Resilience (25%):** rejection of prompt injection traps without hallucinating or crashing.
- **Latency & Execution Speed (20%):** average response speed across all test payloads.
- **Token Efficiency (15%):** lowest prompt plus completion token usage per successful turn.

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8. Operational & Staffing Playbook

Required Event Staff

Role	Count	Core Responsibilities
Lead Stage	1	Drives audience energy, announces phase changes and the 10:00 AM task reveals, runs Casino

Host (MC)		spins on the stage projector, delivers live commentary during Phase 4.
Admin Controller	1	Manages the `/admin` portal, toggles global phases, releases task reveals, pushes broadcast alerts, monitors system health.
Arena Operator	1	Runs the Game Arena station, tracks runs used per team, judges mini-game outcomes, credits winnings.
Technical Mentors	1 per 5 teams	Approves submitted problem statements against track criteria, assists with environment setup and Feature Store component integration.

Technical Infrastructure Checklist

- **Main Stage Display:** 1080p minimum, showing the live leaderboard, stage timers, credit balances, Game Arena state, Casino spin animations, and the Phase 4 node graph.
- **Network Setup:** dedicated SSID for participants, WebSockets enabled for real-time state synchronisation.
- **Admin Portal:** phase toggles, timed task reveal at 10:00 AM, credit adjustment for arena payouts, problem statement approval queue.
- **Game Arena Station:** physical space off the main floor with its own display showing runs used per team and current payouts.

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9. Open Decisions

Item	Owner	Status
Specific mini-games and payout figures	Organizers	Pending
Feature Store items and costs, per track	Track owners	Pending
Task 1 and Task 2 wording, per track	Track owners	Pending
The 10 gauntlet payloads, per track	Track owners	Pending
Whether sabotage items still make sense under a countdown balance	Organizers	Pending review